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CS470 Final Reflection

YouTube link:

<https://youtu.be/VaGuCwngGy0>

Experiences and Strengths

CS 470 has been one of the most valuable courses in my Computer Science program because it provided hands-on experience with cloud computing, serverless architecture, and full-stack application deployment. Throughout this course, I developed and deployed a cloud-based web application using AWS services, allowing me to gain practical experience with technologies commonly used in modern software development environments.

One of the most important skills I learned was how to deploy and manage applications in the cloud. I worked with Amazon S3 to host a static Angular frontend, AWS Lambda to create serverless backend functions, API Gateway to expose REST endpoints, and DynamoDB to store application data. I also gained experience using Docker and Docker Compose to containerize and orchestrate a multi-container application consisting of an Angular frontend, a Node.js backend, and a MongoDB database.

The course strengthened my understanding of cloud architecture, API development, database design, and serverless computing. I learned how individual cloud services work

to create scalable and maintainable applications. I also developed troubleshooting and problem-solving skills while resolving configuration issues involving Lambda functions, API Gateway routing, Docker networking, and DynamoDB integration.

My strengths as a software developer include persistence, adaptability, and the ability to learn new technologies quickly. Throughout my degree program, I balanced full-time employment while completing challenging software engineering coursework. During CS 470, I successfully worked through technical issues and learned how to deploy cloud-based solutions independently.

My previous experience as a Newscast Director has also helped me develop strong troubleshooting, communication, and critical-thinking skills that transfer well into software development and technology roles.

As a result of this course and my Bachelor of Science in Computer Science with a concentration in Software Engineering, I feel prepared to pursue entry-level positions such as Software Developer, Junior Software Engineer, Cloud Support Associate, Cloud Operations Specialist, Technical Support Engineer, Data Engineer Associate, and DevOps or Systems Support roles.

The experience gained in AWS cloud services has provided a strong foundation for future growth in cloud computing and software engineering careers.

Planning for Growth

As this application grows, cloud services provide several options for improving scalability, reliability, and operational efficiency. One of the most effective approaches would be to continue leveraging microservices and serverless architecture. Rather than maintaining a large monolithic application, individual services can be separated into smaller components that perform specific tasks.

This approach simplifies maintenance, improves fault isolation, and allows individual services to scale independently.

AWS Lambda provides significant advantages for future growth because it automatically scales based on demand. As application traffic increases, AWS provisions additional Lambda executions without requiring manual intervention. Error handling can be improved through structured logging, CloudWatch monitoring, retry mechanisms, dead-letter queues, and detailed exception handling within the application code. These features help ensure failures are detected quickly and do not impact the entire system.

Cost prediction is an important factor when planning future expansion. Serverless services such as Lambda use a pay-for-service model where charges are based on actual usage. This can be extremely cost-effective for applications with inconsistent or unpredictable traffic because there are no charges when the application is idle. However, estimating future costs can become more difficult as usage patterns fluctuate.

Containers typically provide more predictable costs because resources are allocated continuously regardless of traffic volume. Organizations can estimate monthly expenses based on the number and size of container instances required.

While containers may be more cost predictable, they often require additional management, monitoring, patching, and scaling configurations compared to serverless solutions.

Several factors would influence future expansion decisions. Serverless architectures offer advantages such as automatic scaling, reduced infrastructure management, faster deployment, and lower operational overhead. However, they can introduce challenges including cold-start latency, execution time limits, and more complex monitoring across distributed services.

Containerized solutions provide greater control over the runtime environment and are well suited for applications requiring long-running processes or

specialized configurations. The tradeoff is increased administrative responsibility and potentially higher costs when resources are underutilized.

Elasticity plays a critical role in cloud decision making because it allows resources to expand and contract automatically based on demand. This prevents overprovisioning while ensuring application performance remains consistent during periods of high traffic. The pay-for-service model further supports business growth by allowing organizations to pay only for the resources they consume rather than purchasing infrastructure upfront.

Overall, cloud computing provides the flexibility, scalability, and cost efficiency necessary to support future growth.

Through this course, I gained a deeper understanding of how cloud services can be used to design modern applications that are reliable, scalable, and capable of adapting to changing business requirements.